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ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY
UNIT CODE: 0612 451 07A

CAT 3

Total marks: 40 **Time: 1 hour 30 minutes**

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Instructions to Candidates

- ✓ Answer ALL questions.
- ✓ Marks are shown in brackets at the end of each question or part.
- ✓ Write clearly and show your working where a calculation is required.
- ✓ Mobile phones and unauthorized materials are not allowed in the examination room.

QUESTIONS

1. Define network testing. **(2 marks)**

2. State two reasons why network testing is important. **(2 marks)**

3. Name the instrument used to measure current without cutting or disconnecting the wire, and state how a voltmeter is connected to the component being tested. **(2 marks)**

4. State the role of Npcap when using Wireshark on a Windows computer. **(2 marks)**

5. Define a VLAN. **(2 marks)**

Question 6 (6 marks)

(a) Define TTL and state what a router does when the TTL of a packet reaches 0. **(2 marks)**

(b) A Windows computer sends a packet with a starting TTL of 128. The packet passes through 9 routers before reaching its destination. Calculate the TTL on arrival and state whether the packet is dropped. **(2 marks)**

(c) Explain how the tracert command uses TTL to discover each router on the path to a destination. **(2 marks)**

Question 7 (6 marks)

A technician runs tracert to a web server and obtains the output below.

```
1   2 ms    1 ms    2 ms   192.168.1.1
2   9 ms    8 ms    9 ms   10.20.0.1
3  26 ms   25 ms   27 ms   41.89.10.5
4 180 ms  178 ms  182 ms   196.201.4.9
5   *       *       *      Request timed out.
6 190 ms  188 ms  191 ms   203.0.113.20
```

(a) Identify the hop where the delay increases sharply and state what this suggests about the network. **(2 marks)**

(b) Give two possible reasons for the asterisks (* * *) shown at hop 5, even though hop 6 replied. **(2 marks)**

(c) Explain why running ping alone would not have shown where the delay occurs. **(2 marks)**

Question 8 (6 marks)

(a) Describe the TCP three-way handshake as seen in Wireshark, stating who sends each packet and the SYN and ACK flag values in each step. **(4 marks)**

(b) Write the Wireshark display filter that would show only:

(i) the first packet (SYN) of each TCP connection request; **(1 mark)**

(ii) ICMP Echo Request packets. **(1 mark)**

Question 9 (6 marks)

A technician is commissioning a new office network and records the three findings below.

Finding	Tool used	Observation
(i)	Voltmeter	A wall socket reads 180 V AC. The expected supply is about 240 V AC.
(ii)	Cable tester	A newly crimped T568B patch cord shows LEDs 1 to 4 and 6 to 8 lighting in order, but LED 5 does not light.
(iii)	Wi-Fi analyser	The signal in the meeting room is measured at -82 dBm.

For each finding, interpret the result and recommend one suitable action. **(2 marks each)**

Question 10 (6 marks)

A school has a Finance department (PC1 and PC2 on ports Fa0/1 and Fa0/2) and a Human Resource department (PC3 and PC4 on ports Fa0/3 and Fa0/4) connected to one Cisco switch. The Finance network is 192.168.10.0/24 and the Human Resource network is 192.168.20.0/24. The school wants the two departments separated.

(a) Write the switch commands to create VLAN 10 named FINANCE and to assign port Fa0/1 to it as an access port. **(3 marks)**

(b) After the VLANs are configured, PC1 (192.168.10.10) cannot ping PC3 (192.168.20.10). Explain why, and describe how communication between the two VLANs can be enabled. **(3 marks)**

END OF QUESTION PAPER

1. Network testing is the process of checking a computer network to ensure that it is working correctly and communicating properly between connected devices

2. To identify faults
To ensure performance

3. Clamp ammeter
A voltmeter is connected in parallel across the component being tested.

4. Npcap allows wireshark to capture and monitor network packets on a computer

5. A VLAN (Virtual local Area Network) is a logical grouping of devices on a network into separate networks regardless

6 TTL (Time to live) - is the number of hops a data packet is allowed to pass through before it is discarded.
When the TTL reaches 0, the router discards the packet and sends an ICMP Time Exceeded message to the source device

b) Each router reduces the TTL by 1
128 - 9 = 119
The packet is not dropped, because the TTL has not reached 0 when the packet is usually dropped

c) The tracert command sends packets with increasing TTL values, starting from 1.
Each router that reduces the TTL to 0 discards the packet and sends an ICMP Time Exceeded message, allowing tracert to identify the router's IP address.

7a) Hop 4 - the delay increases at hop 3 26ms hop 3 to 150 ms hop 4.
This suggests network congestion or a delay along that part of the network path.

7b) Firewall filtering - the router at hop 5 may block ICMP requests or responses.
Request timeout - the router at hop 5 may be too busy or configured not to respond within the allowed time.

c) Ping only measures the round-trip time between the source computer and

destination.

8a) The TCP three-way handshake in wireshark

SYN - the client sends a SYN packet to the server to request a connection. SYN=1 Ack=0

SYN-Ack - the server replies with a SYN-Ack packet. SYN=1, Ack=1

Ack - the client sends an Ack packet to confirm the connection. SYN=0, Ack=1

This establishes a TCP connection between the client and the server.

b) i) tcp.flags.syn == 1 && tcp.flags.ack == 0

ii) icmp.type == 8

9.) Voltage.

The voltage supply is too low.

~~Actions that can be taken are~~

The action that can be taken is to report the fault to an electrician to inspect the socket and electrical supply.

i) Cable tester

~~There~~ Cause - there is an open circuit or broken connection in pin 5

The action to be taken is to re-crimp the RJ45 connector, ensuring wire 5 is correctly inserted.

ii) Wi-Fi analyser.

~~Cause~~ The Wi-Fi signal is weak and may cause slow speeds

The action to be taken is to move the access point closer to the meeting room.

10a) ~~enable~~. Commands to create VLAN 10 named Finance.

enable

configure terminal

vlan 10

name FINANCE

exit

interface fa0/1

Switchport mode access

Switchport ~~mode~~ access vlan 10

end. ✓

b) i) VLANs seperate network traffic onto different broadcast domains. Devices in different VLANs and IP subnets cannot communicate directly through a layer 2 switch.

ii) To enable communication one must:
Configure inter-VLAN routing using a router or a layer 3 switch.
Configure the appropriate IP addresses, subnet masks and default gateways for both VLANs.

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